ASM 591 Homework Assignment: Weather Station Dashboards (Grafana + PostgreSQL)

Overview

In this assignment, you will load `weather_station.csv` into a PostgreSQL database, connect the database to Grafana, and build dashboards from multiple weather and soil measurements. Please see SPRING Portal Weather Dashboard for some examples before completing your tasks. You will:

1. Inspect a CSV file and infer column meanings and data types.
2. Write a SQL `CREATE TABLE` statement (you may use generative AI tools such as ChatGPT).
3. Use `csv_to_sql.py` to generate a bulk `INSERT` script and populate the table.
4. Add PostgreSQL as a Grafana data source and create panels for your assigned task set.

Files and tools

- **Dataset:** `weather_station.csv`
- **Script:** `csv_to_sql.py` (generic CSV → bulk SQL generator)
- **Database:** PostgreSQL (TimescaleDB)
- **Dashboard:** Grafana

Learning objectives

By the end of this assignment, you should be able to:

- choose appropriate SQL types for time-stamped multi-sensor data,
- bulk-load CSV data into a relational database,
- write SQL queries for time series, bar charts, stat/gauge panels, and aggregations,
- build dashboards that respond to Grafana's time range selector.

Getting connected

Connecting to the PostgreSQL database (Beekeeper Studio)

Use Beekeeper Studio to connect to the class PostgreSQL database.
Connection details:

- **Host:** `aglab`
- **Port:** `5432`
- **Database:** `aglab`
- **Username:** `studentX` (`X` will be provided by the instructor)
- **Password:** (provided by the instructor)

Steps:

1. Open Beekeeper Studio and click **New Connection**.
2. Choose **PostgreSQL**.
3. Enter the connection details above.
4. Click **Test Connection**, then click **Connect**.

Connecting to Grafana at `aglab:3000` (Tailscale)

Grafana is available at `http://aglab:3000`. You must join the course Tailscale network to access it.

Step 1: Join the course Tailscale network

1. Use the instructor-provided invite (or the invite sent to your email) and join the tailnet.
2. Confirm that Tailscale shows you are **connected**.

Step 2: Open Grafana

- In a browser, go to: `http://aglab:3000`
- Log in with:
  - **Username:** `studentX`
  - **Password:** (same as your database password)

Part A: Create the SQL table (you may use ChatGPT)

Write a `CREATE TABLE` statement for a table named `weather_station`. Your table must:

- store timestamps correctly (timezone-aware),
- store the device identifier,
- store measurements using sensible numeric types,
- enforce a sensible primary key.

Part B: Convert CSV to a bulk INSERT and load the data

1) Generate a bulk SQL file

Use `csv_to_sql.py` to generate a **single bulk INSERT** file.

2) Load the data into PostgreSQL

- Connect to your database using Beekeeper Studio.
- Open the SQL query editor and paste the contents of `weather_station_bulk.sql` and execute.

3) Verify the data

```
SELECT COUNT(*) FROM weather_station;
```

Part C: Connect Grafana to PostgreSQL

1. In Grafana: **Connections** → **Data sources** → **Add data source**.
2. Select **PostgreSQL**.
3. Fill in the connection settings:
   - **Host:** `aglab-timescaledb:5432`
   - **Database:** `aglab`
   - **User:** `studentX`
   - **Password:** (provided by instructor)
   - **TLS/SSL Mode: Disable**
4. Click **Save & Test**.

Part D: Dashboard tasks (assigned per student)

Each student will build a dashboard named **Weather Station Assignment – StudentX** using SQL queries and complete **their assigned task set**. Each panel must have:

- a clear title,
- correct units in the title or axis,
- a query that respects the dashboard time range selector (`$__timeFilter(ts)`).

Note (important): The dataset is not live. If your time range is set to "Last 6 hours" the panels may appear empty. Choose a time range that includes the CSV timestamps.
Note 2: Do not forget to switch from `Builder` to `Code` editor before writing your queries.

Student 3 task set: Temperature + Humidity

Create 4 panels:

1. **Air temperature (Gauge):** most recent `air_temp_c`.
2. **Air temperature (Time series):** `air_temp_c` over time.
3. **Relative humidity (Time series):** `rel_hum` over time.
4. **Daily average relative humidity (Bar Graph):** 1-day buckets of `AVG(rel_hum)`.

Student 5 task set: Soil + Water Potential

Create 4 panels:

1. **Soil temperature (Stat):** most recent `temp_c`.
2. **Soil water potential (Time series):** `poten_kpa` over time.
3. **Soil temperature (Time series):** `temp_c` over time.
4. **Soil temperature by depth (Time series, multiple lines):** time series grouped/legend by `depth_cm`.

Student 6 task set: Wind

Create 4 panels:

1. **Wind speed (Gauge):** most recent `wind_sp_ms`.
2. **Wind speed (Time series):** `wind_sp_ms` over time.
3. **Wind gust (Time series):** `wind_sp_gust_ms` over time.
4. **Wind components (Time series, multi-line overlay):** overlay `wind_north_sp_ms` and `wind_east_sp_ms`.

Student 7 task set: Radiation + Pressure

Create 4 panels:

1. **Solar radiation (Stat):** most recent `radi_w_m2`.
2. **Solar radiation (Time series):** `radi_w_m2` over time.
3. **Atmospheric pressure (Time series):** `atmos_pres_kpa` over time.
4. **Daily max solar radiation (Bar Graph):** 1-day buckets of `MAX(radi_w_m2)`.

Student 9 task set: PAR + Vapor Pressure

Create 4 panels:

1. **PAR (Gauge):** most recent `par_umol_m2_s`.
2. **PAR (Time series):** `par_umol_m2_s` over time.
3. **Vapor pressure (Time series):** `vapor_pres_kpa` over time.
4. **Radiation vs PAR (Time series, multi-line overlay):** overlay `radi_w_m2` and `par_umol_m2_s` in one panel.

What to submit

Submit the following on Brightspace under the assignment:

1. First, please specify your `studentX` username and the assigned task in your report.
2. **Part A (Schema):** your final `CREATE TABLE` SQL (copy/paste).
3. **Part B (Loading):** the exact `csv_to_sql.py` command you ran, and the verification query result (e.g., `SELECT COUNT(*) ...;`).
4. **Part D (Dashboard queries):** the SQL query for each of your 4 assigned panels.
5. **Screenshots:** one screenshot per panel showing the panel title and the visualization.

Grading checklist

Item	Criteria	Points
Schema	Correct types; sensible primary key; table created successfully	20
Load	Bulk insert generated and data loaded; verification query shown	20
Dashboards	4 correct panels for your assigned task set	40
Screenshots	1 screenshot per panel showing title and visualization	20